European Research Council
Established by the European Commission

Structured Interactive Perception and Learning for Holistic Robotic Embodied Intelligence

Fact Sheet

Project Information

SIREN

Grant agreement ID: 101163933

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10.3030/101163933

EC signature date
23 September 2024

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1 June 2025

End date
31 May 2030

Funded under
European Research Council (ERC)

Total cost
€ 1 499 738,00

EU contribution
€ 1 499 738,00

Investment in EU policy priorities

Digital agenda
Clean air
Artificial Intelligence
Climate action
Biodiversity

Coordinated by
TECHNISCHE UNIVERSITAT DARMSTADT
Germany

Project description

A holistic approach to intelligent adaptation

Robots have advanced in learning capabilities due to powerful neural models and vast data sets. However, critical questions remain: Do large architectures and extensive data truly foster robotic intelligence that mirrors human intuition? Moreover, how can we enhance robot learning systems to thrive in dynamic, real-world environments? To answer these questions, the ERC-funded SIREN project will present a holistic view of robot learning, integrating robots and their surroundings as a cohesive system. By investigating the action-perception cycle and leveraging graph neural networks, SIREN aims to develop adaptive robots capable of executing complex tasks in unstructured environments. This paradigm shift promises to pave the way for continuous, evolution-based learning in robotics.

Objective

Robot learning has made remarkable strides thanks to high-capacity neural models and extensive datasets. However, there are persisting research questions concerning large-scale robot learning models: are massive architectures and data needed for achieving robotic embodied intelligence to solve tasks intuitive to humans? And how can we make substantial progress toward robust and adaptive robot learning systems to operate in the dynamic real world? I posit that these open problems stem from overlooking the underlying principles and structure that govern the intricate robot-environment interaction and evolution.

SIREN addresses these pressing issues by proposing a unique systemic view of robot learning through the holistic representation of robot and environment as an integrated system. To achieve this, we will unveil key properties of the action-perception cycle for developing embodied intelligence by studying the intertwined flow of information and energy within the components of the holistic system. For that, we propose a framework that pioneers information-driven and physics-aware objectives that encompass the learning from embodied multisensorial streams of a modular graph representation of the robot-environment system and its dynamics, backed by the versatility of graph neural networks, allowing for modular uncertainty estimation to promote robustness. Eventually, we will yield resilient dynamics for training uncertainty-aware, composable skills to adapt to new tasks. SIREN's breakthroughs will enable robots, like humanoid mobile manipulators, to merge in unstructured, human-like settings and perform challenging tasks that require smooth and efficient perception-action coordination, balancing generalization and robustness in the face of inevitable real-world uncertainties. Our paradigm shift opens avenues for future groundbreaking research rooted in SIREN's impacts toward continuous robot learning systems that are integrated and evolve with their environment.

Fields of science (EuroSciVoc)

engineering and technology > electrical engineering, electronic engineering, information engineering > electronic engineering > **robotics**

Keywords

robot learning

multisensory perception

graph neural networks

mobile manipulation

Programme(s)

HORIZON.1.1 - European Research Council (ERC)

MAIN PROGRAMME

See all projects funded under this programme

Topic(s)

ERC-2024-STG - ERC STARTING GRANTS

See all projects funded under this topic

Funding Scheme

HORIZON-ERC - HORIZON ERC Grants

See all projects funded under this funding scheme

Call for proposal

ERC-2024-STG

See all projects funded under this call

Host institution

TECHNISCHE UNIVERSITAT DARMSTADT

Net EU contribution

€ 1 499 738,00

Total cost

€ 1 499 738,00

Address

KAROLINENPLATZ 5
64289 DARMSTADT
Germany

Region

Hessen > Darmstadt > Darmstadt, Kreisfreie Stadt

Activity type

Higher or Secondary Education Establishments

Links

Contact the organisation   Website
Participation in EU R&I programmes
HORIZON collaboration network

Beneficiaries (1)

TECHNISCHE UNIVERSITAT DARMSTADT

Germany

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